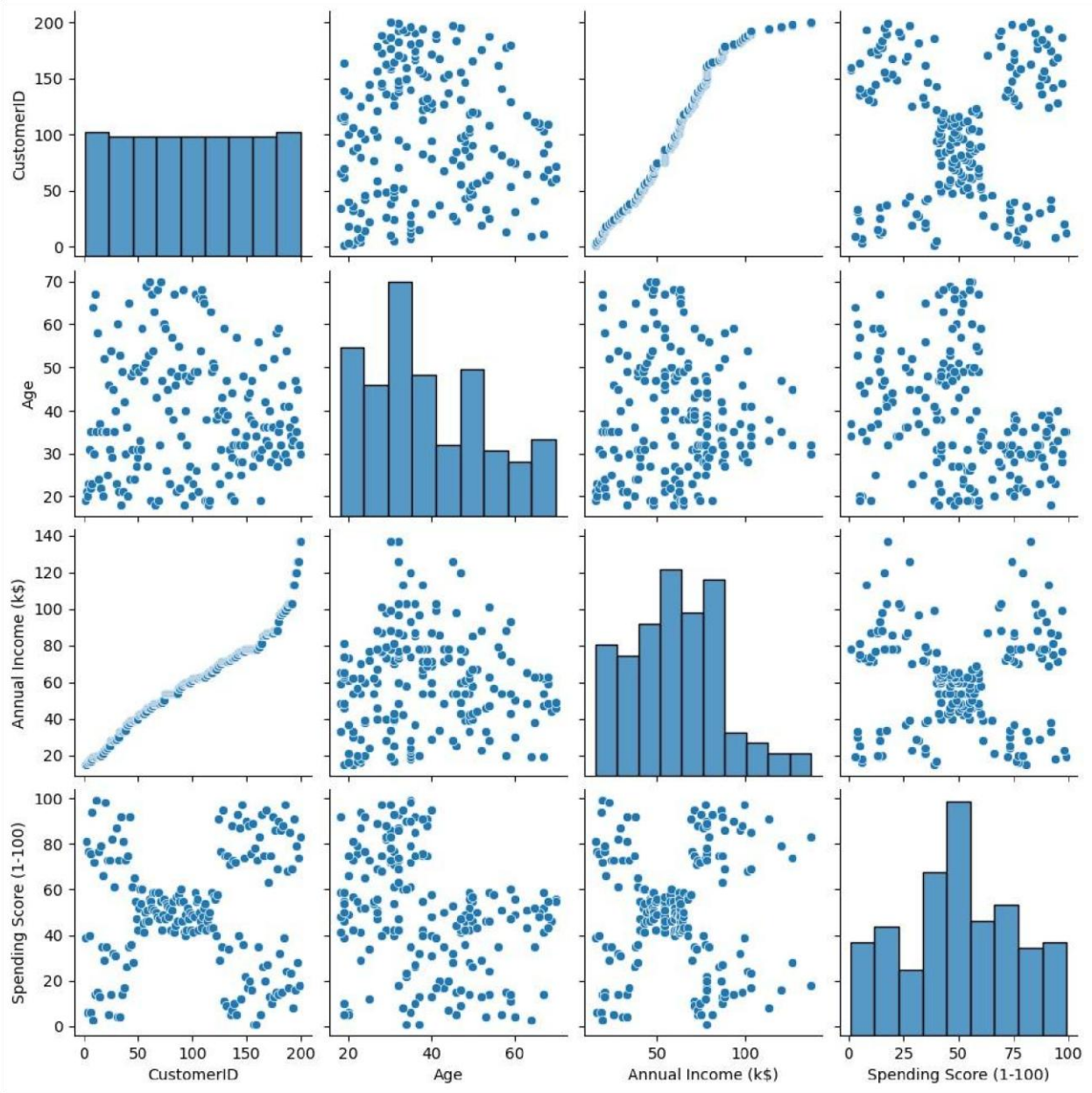
Name:Tharunya BS Register Number: 250701786 Experiment Name:K-Means Clustering

```
import numpy as np import
pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns %matplotlib
inline
df=pd.read_csv('Mall_Customers.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----  0
     CustomerID               200 non-null    int64
1    Gender                   200 non-null    object
2    Age                      200 non-null    int64
3    Annual Income (k$)       200 non-null    int64  4   Spending
Score (1-100)  200 non-null    int64
dtypes:  int64(4),  object(1)  memory
usage: 7.9+ KB df.head()
```

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
sns.pairplot(df)
<seaborn.axisgrid.PairGrid at 0x1f67ce53230>
```




```
features=df.iloc[:,[3,4]].values

from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)

C:\Users\swath\anaconda3\Lib\site-packages\sklearn\cluster\
_kmeans.py:1419: UserWarning: KMeans is known to have a memory leak on
Windows with MKL, when there are less chunks than available threads.
You can avoid it by setting the environment variable
OMP_NUM_THREADS=1.   warnings.warn(
```

```
KMeans(n_clusters=5)
```

```
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
```

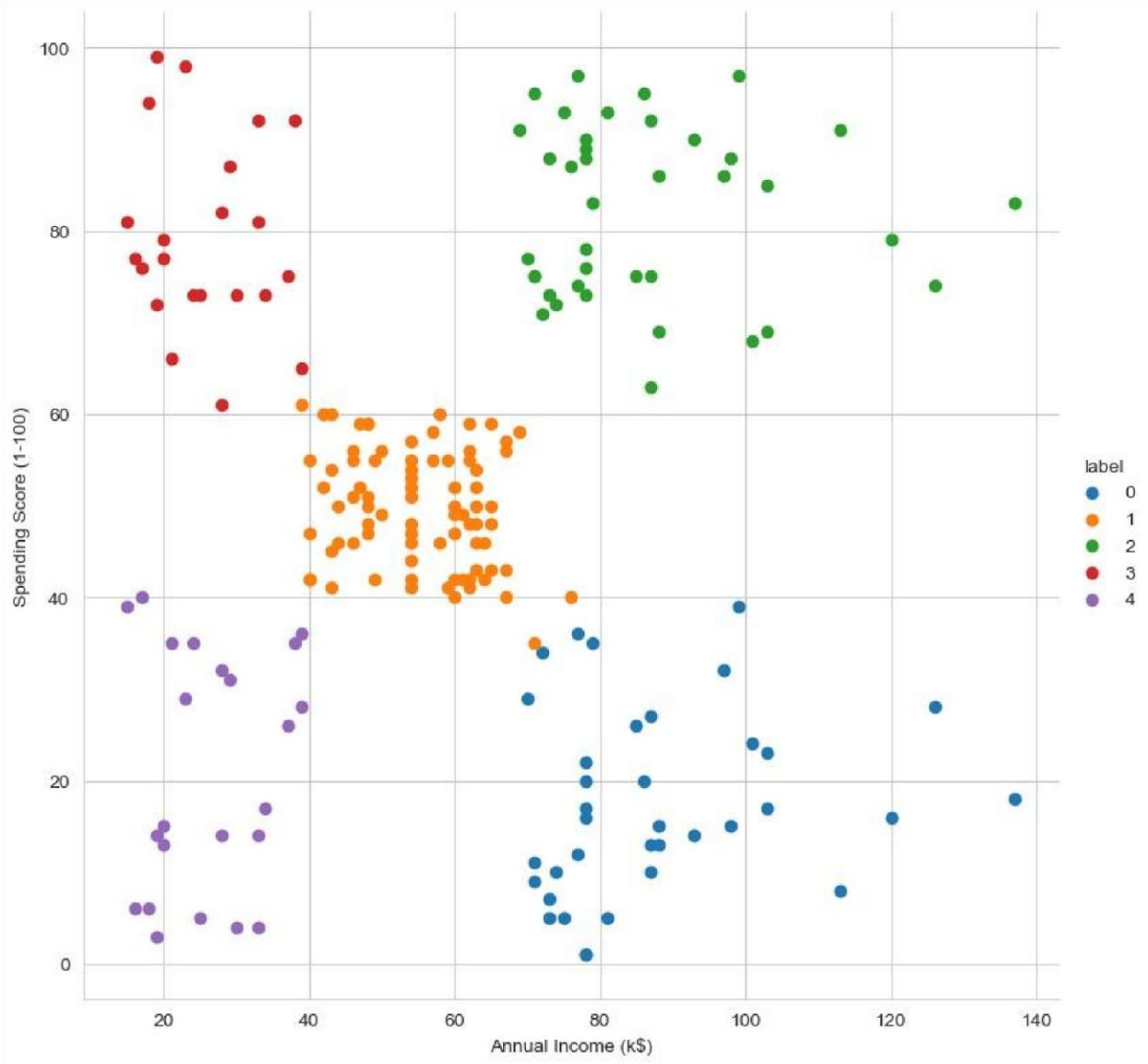
```
C:\Users\swath\AppData\Local\Temp\ipykernel_5828\470183701.py:2:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#
returning-a-view-versus-a-copy
Final['label']=model.predict(features)
```

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	3
2	16	6	4
3	16	77	3
	17	40	4

4

```
sns.set_style("whitegrid") sns.FacetGrid(Final,hue="label",height=8)
\
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend(); plt.show()
```




```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
model.fit(features_el)
wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

```
  Cell In[11], line 5
    model=KMeans(n_clusters=i)
    ^
IndentationError: expected an indented block after 'for' statement on
line 4
```